



Web application to Detect Electrical Signal Problems using Machine Learning

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Course: CSIS 4495

Individual Project

GitHub: https://github.com/CoderOphilia/F26_4495_S2_OphiliaF.git

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Introduction

Problem Definition

Electricity is supposed to flow smoothly; however, glitches can occur due to sudden voltage drops, spikes, cutouts, or distortion, known as power quality disturbances. According to a European survey, they cost businesses about 150 billion Euros every year (Jain et al., 2025). Hospitals, factories, data centers, and semiconductor plants are especially affected because even a split-second glitch can break sensitive equipment or ruin an entire production run. This research will focus on five types of disturbances named: Sag, Swell, Interruption, harmonic distortion and normal.

Preliminary Research

This project is built on three real, peer-reviewed research papers.

1. Jain et al. (2025), published in Computers and Electrical Engineering, is the main base paper. They reviewed over 100 machine learning methods for detecting power quality disturbances and ran a practical case study comparing different classifiers. Their best result was 96.6% accuracy using a method called Bagged Trees. This project will independently reproduce their approach in Python and compare results directly to their published numbers.
2. Abu Nassar & Morsi (2024), published in Electric Power Systems Research, show that power quality disturbances in smart electrical substations can look like cyberattacks, making detection even more important. Their work motivates why reliable detection under noisy conditions matters beyond just equipment protection.
3. Zhao et al. (2021), published in IEEE Transactions on Power Electronics, provide a broad overview confirming that machine learning is now the dominant approach for detecting faults and disturbances in power electronic systems, with supervised classification being used in 91% of cases.

Existing research has already built systems that can automatically detect these disturbances using machine learning. However, there is an important gap. A system that works perfectly on clean data might fail badly on noisy real-world data. My applied research aims to clarify two questions. They are:

- Can I build a machine learning classifier that correctly identifies which type of disturbance is present in a voltage signal, matching the accuracy reported by Jain et al. (2025) in their published research?
- How does that accuracy change when I add realistic noise to the signal? At what noise level does the system start to struggle?

I believe these questions matter practically because the answer tells engineers whether they need an expensive, high-precision sensor or whether a cheaper, noisier one will still do the job reliably.

Hypothesis and Potential Benefit

1. Features extracted from the frequency domain (FFT) will hold up better under noise than raw time-domain features.
2. Accuracy will drop as noise increases
3. Harmonic distortion will be easier to detect under noise than sag or swell, because its signature shows up clearly in the frequency spectrum.

This project gives an open-source answer to the question “how much noise can this system tolerate?”, something the current published literature does not provide in a standardized way.

Proposed Research Methodology

Designing Solution

The deliverable is a full-stack web application. The name of the application is still a work in progress. The steps to get the result are as follows:

1. Users must open the app and upload the CSV file containing the voltage recording
2. The app will display a live chart of the waveform so that the user can see what the signal looks like
3. There will be a slider for the user to adjust the noise level and update the waveform chart in real time when the noise is added
4. There will be a classify button. When the user clicks the classify button, the app will send a signal and be run by the ML model
5. The app should display the result, such as Voltage sag or whatever the result is, with the confidence level

Data Analysis Technique

- Data collection: I will not be using any real electrical equipment for my research given the time frame to complete the course, and it is hard to find a lab with all the equipment. I will try to create voltage signals synthetically using Python and using the mathematical formulas defined in the IEEE standards, which is the approach used by Jain et al. (2025) in their case study. These synthetic signals will be validated by visually comparing them to real signals from the publicly available IEEE DataPort dataset.
- Feature Extraction: Each signal will be summarized into a small set of numbers that describe it, like its average voltage, its peak, and which frequencies are strongest (FFT). These numbers are what the model learns from, not the raw waveform.

- Model Training: Two classifiers will be trained and compared. A Random Forest and a Support Vector Machine. Both are used by Jain et al. (2025), so results can be directly compared.
- Noise-robustness experiment: The trained models are then tested on noisy versions of the same signals at four noise levels (Clean, 40dB, 30dB, 20dB SNR). The results are plotted as accuracy curves to show exactly where each model starts to struggle

Expected Results

- On clean signals (no noise): The classifier is expected to reach around 96% accuracy, matching the benchmark from the Jain et al. (2025) research paper.
- As noise increases: Accuracy is expected to drop. The key finding will be identifying the exact point where accuracy falls below 90%; that's the practical "break point" that tells engineers how noisy a sensor can be before the system stops working reliably.
- I aim to pinpoint exactly where and how the sensor starts to fail. Anyone can use it through a web interface without needing a signal processing background. A real-world application can be electrical utilities like BC Hydro monitoring grid health, hospitals protecting life-critical equipment, data centers preventing server crashes, and transit systems like SkyTrain getting instant fault alerts.

Project Planning and Timeline

This is an individual project. All work is done by Ophilia Fradarick (ID: 300386351). Dates are mapped to the course assessment schedule.

Dates	Work Being Done	Deliverable Due
Sept 15–28	Topic confirmed; GitHub repo set up; signal generator script written and tested in Python	Proposal + Repo submitted (Sep 28)
Sept 29 – Oct 12	Full synthetic dataset generated; exploratory data analysis done; Flask project skeleton created	Progress Report 1 (Oct 12)
Oct 13–26	Feature extraction coded; Random Forest and SVM trained on clean signals; basic front-end, connected to Flask back-end	Mid-Course Demo + Report (Oct 26): working app with signal display
Oct 27 – Nov 16	Noise slider added to app; noise experiments run at 40dB, 30dB, 20dB; accuracy curves plotted; results compared to Jain et al. benchmarks	Progress Report 2 (Nov 16): app shows live noise effect on waveform

Nov 17–23	Classification result panel added to app (shows predicted class + confidence score); SQLite database connected to log all results	Progress Report 3 (Nov 23): app classifies and stores results
Nov 24–30	History panel added (shows past classifications from database); full end-to-end app testing; final bug fixes	Progress Report 4 (Nov 30): fully working app
Dec 1–7	Final report written; GitHub repo cleaned up with README; app demo rehearsed	Final Report + Working App (Dec 7)
Dec 9	Live demo of the full app in class; present noise-robustness findings	Defense Presentation

Technologies will be used

Platform	Runs locally on any laptop	No cloud server required for this project scope
Programming Language	Python 3	Standard for machine learning and signal processing; same as Jain et al. (2025)
Machine Learning Libraries	NumPy, SciPy, PyWavelets, scikit-learn	Feature extraction (FFT, DWT) and classifier training (Random Forest, SVM)
Back-End	Flask (Python)	Lightweight web server connecting the trained ML model to the front-end website
Front-End	HTML5, CSS3, JavaScript,	Interactive website with waveform chart, noise level slider, and classify button
Database	SQLite	Stores each classification result (signal type, noise level, predicted class, confidence score) for the history panel

Project Contract

I, Ophilia Fradarick (Student ID: 300386351), agree to complete this project individually for CSIS 4495 – Section 002, Fall 2026. I will follow the scope, methodology, and timeline

described in this proposal. I understand that a fully working web application is required to receive a non-zero final grade, and that I must attend all three mandatory in-person check-in sessions.

Signature: Ophilia Fradarick

Date: September 25, 2026

AI Use Section

All AI tools used in preparing this proposal are listed below, as required by the CSIS 4495 Generative AI Policy. The full prompt history is in the Appendix.

AI Tool	Version	What it was used for	Value I added over the AI
Claude	Claude Sonnet – Pro version	Verified research papers I uploaded myself and helped me understand the engineering terms.	I chose and uploaded all three research papers myself. I directed all topic and scope decisions. I found the research papers after the meeting I had with the Douglas College librarian.

Appendix (Prompt History Placeholder)

1. September 9, 2026: I am a student who is new to research. Where should I start? How do I write a research proposal?
2. September 18, 2026: Is it feasible to research cyberattack detection in smart grids within one semester?
3. September 20, 2026: Is there any application of this research in BC Hydro or TransLink?
4. September 23, 2026: Can you show me what the IEEE DataPort dataset looks like?
5. September 24, 2026: I want to create something realistic that will improve my SQL skills and data analysis skills.
6. September 24, 2026: Can I create an app or chatbot system for this? My professor wants a product that incorporates front-end and back-end, a full-stack operation project.
7. September 25, 2026: Give me all the prompts I asked for, with dates.

Work log

Date	Hours	What was done
September 9, 2026	2.5	Brainstormed project ideas during the first lecture, discussed them with the professor, and received approval to proceed with the project.

September 16, 2026	0.5	Spoke with an engineering manager about becoming my industry partner. However, their area of expertise is different from the focus of my project.
September 18, 2026	1	Met with Kalachinskaya, Olga, Faculty Subject Librarian for ENGR, MATH, CSIS, CMPT, VTEC, and VOAC, to discuss how to find research papers to support my Applied Research Project.
September 18, 2026	0.25	Created GitHub repo (F26_4495_S2_OphiliaF); added professor kandhadaip@douglascollege.ca as collaborator
September 20, 2026	3	Found three research papers that can be used to support my project.
September 23, 2026	1	In-person check-in with professor; confirmed project topic and scope
September 25, 2026	4	Wrote and edited the project proposal, consolidated the content, and simplified the wording to make the proposal clearer and easier to understand.
September 26, 2026	0.25	Meeting with Professor
September 28, 2026	1	Finalizing the proposal and submitting it

References

- [1] S. Jain, A. Satsangi, R. Kumar, D. Panwar, and M. Amir, "Intelligent assessment of power quality disturbances: A comprehensive review on machine learning and deep learning solutions," *Computers and Electrical Engineering*, vol. 123, p. 110275, 2025.
<https://doi.org/10.1016/j.compeleceng.2025.110275>
- [2] A. Abu Nassar and W.G. Morsi, "Detection of Cyber-Attacks and Power Disturbances in Smart Digital Substations using Continuous Wavelet Transform and Convolutional Neural Networks," *Electric Power Systems Research*, vol. 229, p. 110157, 2024.
<https://doi.org/10.1016/j.epsr.2024.110157>

[3] S. Zhao, F. Blaabjerg, and H. Wang, "An Overview of Artificial Intelligence Applications for Power Electronics," *IEEE Transactions on Power Electronics*, vol. 36, no. 4, pp. 4633–4658, 2021. <https://doi.org/10.1109/TPEL.2020.3024914>

[4] IEEE Standard 1159-2019, "IEEE Recommended Practice for Monitoring Electric Power Quality," IEEE, 2019.